Status	Finished
Started	Thursday, 1 October 2026, 4:12 PM
Completed	Thursday, 1 October 2026, 4:21 PM
Duration	9 mins 39 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Hashing
c. Binary search
d. Linear search ✓

Question **2**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 1000
c. 500
d. 100

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It can only be applied to large lists
c. It is slower than linear search for small lists
d. It does not work with negative numbers

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 89 and 99

c. 89 and 94

d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is the last element in the array

Question **6**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list

b. Binary search can find multiple instances of the target element

c. Binary search works on unsorted lists

d. Binary search has a lower time complexity on large, sorted lists ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓

b. By using a hash function

c. By starting from the first element

d. By comparing each element sequentially

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Sorting the list before searching
d. Checking each element sequentially ✓

Question **9**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Hash search
c. Both (1) & (3)
d. Binary search

Question **10**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. n-1

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. At the beginning of the list
c. In the right sublist
d. In the entire list

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓
d. The list must be sorted

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The search is considered unsuccessful ✓
d. The first element is returned

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. There must be a mechanism to access middle element directly